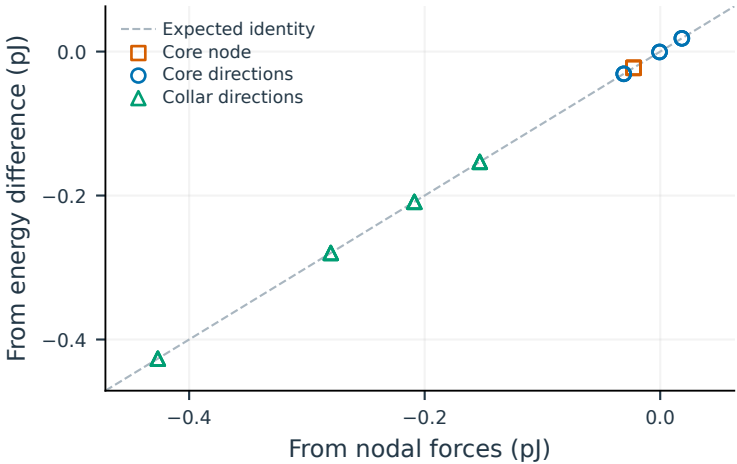
Energy–force check

Archived first-variation probes

$$\mathrm{d}U/\mathrm{d}\eta = -\sum_i \mathbf{f}_i \cdot \mathbf{v}_i$$



All 24 saved comparisons; coincident points overlap.

Energy and forces agree

8 directions × 3 steps; max error 9.49e-06

All 24 saved comparisons are plotted; markers can overlap.
Particle-position glyphs are schematic, not new simulation states.